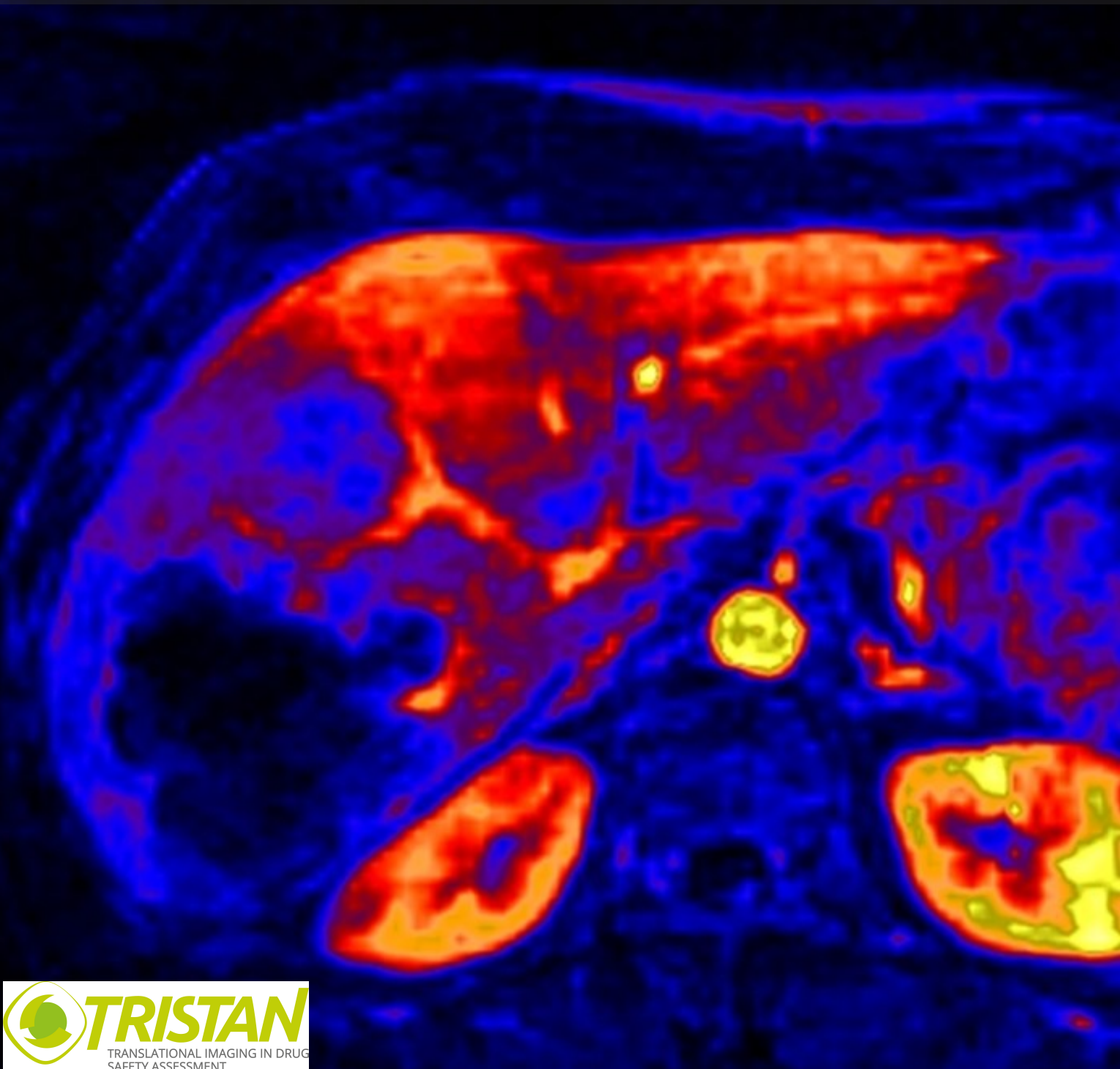


Leeds pilot study

rifampicin (key results)

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by

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Report compiled by:	Steven Sourbron
Institute:	University of Sheffield
Department:	Section of Medical Imaging and Technologies
Email:	s.sourbron@sheffield.ac.uk
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Key results

1.1. Data summary

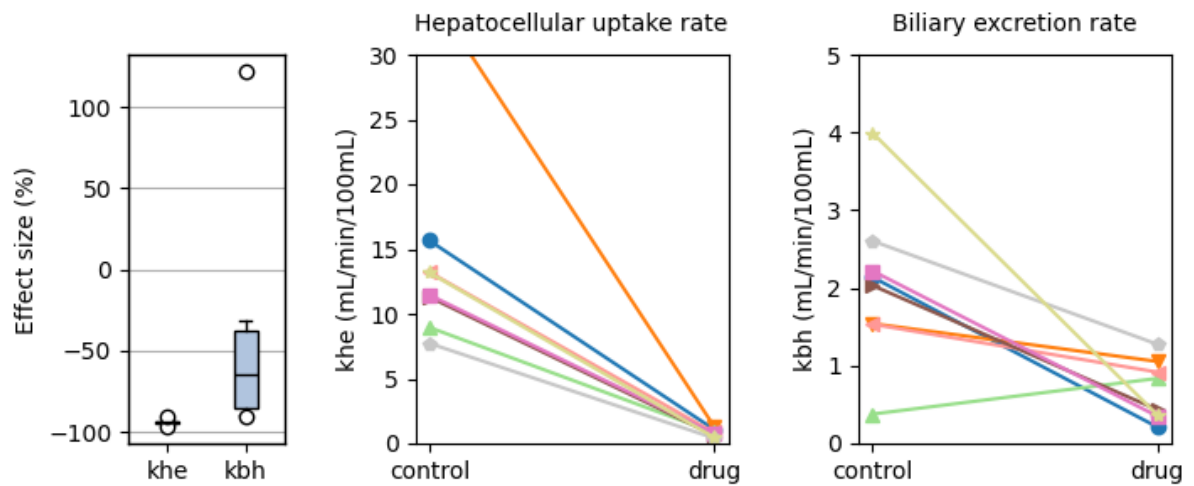


Figure 1.1: Effect size (%) on hepatocellular uptake (k_{he} , left) and biliary excretion (k_{bh} , right) of gadoxetate. The boxplot shows median, interquartile range and 95 percent range. The line plots show individual values for hepatocellular uptake (k_{he} , middle) and biliary excretion (k_{bh} , right) of gadoxetate of the control (left of plot) and treatment (right of plot). Grey lines are healthy controls with rifampicin injection.

	parameter	count	mean	std	min	25%	50%	75%	max
	khe effect size (%)	8.0	-94.0	1.9	-96.3	-95.0	-93.9	-93.2	-90.5
	khe control (mL/min/100cm ³)	8.0	14.55	8.59	7.72	10.71	12.32	13.83	34.88
	khe drug (mL/min/100cm ³)	8.0	0.81	0.31	0.42	0.61	0.8	0.91	1.34
	kbh effect size (%)	8.0	-43.1	70.4	-90.8	-85.6	-64.7	-38.0	121.3
	kbh control (mL/min/100cm ³)	8.0	2.05	1.03	0.38	1.54	2.08	2.31	3.99
	kbh drug (mL/min/100cm ³)	8.0	0.68	0.39	0.21	0.36	0.64	0.95	1.28

Table 1.1: Effect size and absolute values of hepatocellular uptake (k_{he}) and biliary excretion (k_{bh}) of gadoxetate

1.2. Liver biomarkers

Biomarker	p-value	Bayes Factor	Odds Ratio
AUC for CI (0-35min)	0.0001	264.17	1017.19
AUC for CI (0-inf)	0.00199	23.64	50.25
Biliary excretion rate	0.01533	4.68	24.4
Biliary tissue excretion rate	0.01661	4.4	20.69
Final biliary excretion rate	0.07399	1.42	6.03
Final hepatocellular uptake rate	0.02396	3.31	13.0
Gut dispersion	0.37833	0.48	0.42
Gut mean transit time	0.30326	0.54	0.32
Hepatocellular mean transit time	0.16657	0.8	0.21
Hepatocellular tissue uptake rate	0.02861	2.89	12.13
Hepatocellular uptake rate	0.00232	20.86	60.47
Initial biliary excretion rate	0.12207	0.99	4.95
Initial hepatocellular uptake rate	2e-05	1054.0	9919.24
Liver T1-MOLLI at 45min	0.0	6244.03	0.0
Liver T1-MOLLI at baseline	0.3834	0.48	0.73
Liver T1-MOLLI at scan 2	0.00215	22.19	0.06
Liver extracellular volume fraction	0.03628	2.42	0.1
Liver plasma clearance	0.00187	24.85	73.73
RE for R11 at 20min	7e-05	376.64	1013.36
RE for SI at 20min	8e-05	342.72	813.84

Table 1.2: Results of a pairwise comparison testing for differences in liver biomarkers between control and treatment. The results are ranked by their p-value, with most significant differences at the top of the list.

Biomarker	Units	control	drug	change (%)
AUC for CI (0-35min)	mM*sec	177.0 (36.0)	32.8 (5.8)	-80.4 (4.1)
AUC for CI (0-inf)	mM*sec	562.0 (190.0)	129.0 (25.0)	-73.9 (5.8)
Biliary excretion rate	mL/min/100cm3	2.05 (0.71)	0.682 (0.27)	-43.1 (49.0)
Biliary tissue excretion rate	/min	0.0272 (0.0077)	0.0116 (0.005)	-26.3 (70.0)
Final biliary excretion rate	mL/min/100cm3	1.96 (0.79)	1.04 (0.46)	-13.9 (61.0)
Final hepatocellular uptake rate	mL/min/100cm3	18.4 (12.0)	0.962 (0.26)	-93.0 (2.3)
Gut dispersion	%	61.5 (15.0)	69.3 (4.6)	28.7 (40.0)
Gut mean transit time	sec	33.8 (13.0)	43.3 (7.5)	72.3 (72.0)
Hepatocellular mean transit time	min	60.0 (51.0)	131.0 (64.0)	270.0 (200.0)
Hepatocellular tissue uptake rate	/min	1.77 (1.2)	0.0398 (0.012)	-96.3 (1.4)
Hepatocellular uptake rate	mL/min/100cm3	14.5 (6.0)	0.806 (0.21)	-94.0 (1.3)
Initial biliary excretion rate	mL/min/100cm3	2.62 (0.97)	1.27 (1.1)	-36.7 (52.0)
Initial hepatocellular uptake rate	mL/min/100cm3	10.7 (1.9)	0.649 (0.2)	-93.6 (2.3)
Liver T1-MOLLI at 45min	sec	0.537 (0.036)	0.727 (0.041)	35.7 (5.9)
Liver T1-MOLLI at baseline	sec	0.795 (0.037)	0.805 (0.041)	1.27 (2.7)
Liver T1-MOLLI at scan 2	sec	0.669 (0.041)	0.759 (0.038)	13.8 (6.3)
Liver extracellular volume fraction	mL/100cm3	13.1 (5.4)	21.1 (2.9)	196.0 (230.0)
Liver plasma clearance	L/min	0.13 (0.051)	0.00738 (0.0015)	-93.7 (1.6)
RE for R11 at 20min	%	63.7 (14.0)	8.8 (1.5)	-85.7 (1.8)
RE for SI at 20min	%	53.4 (12.0)	7.83 (1.8)	-84.9 (2.5)

Table 1.3: Mean values along with their 95 percent confidence intervals for all liver biomarkers of the control and treatment visit. The last column shows the relative change at the treatment visit. The results are ranked by their p-value, with most significant differences at the top of the list.

1.3. Systemic biomarkers

Biomarker	p-value	Bayes Factor	Odds Ratio
AUC for Cb (0-35min)	0.05248	1.83	0.34
AUC for Cb (0-inf)	0.05911	1.68	0.22
Body extraction fraction	0.04209	2.16	7.85
Cardiac output	0.2363	0.64	0.4
Heart-lung dispersion	0.07318	1.43	0.18
Heart-lung mean transit time	0.86741	0.34	1.14
Organs blood mean transit time	0.49445	0.42	0.66
Organs extraction fraction	0.00036	94.79	15.81
Organs extravascular mean transit time	0.25136	0.61	0.43
RE for R1b at 20min	0.0241	3.3	0.22
RE for Sb at 20min	0.71646	0.36	0.7

Table 1.4: Results of a pairwise comparison testing for differences in systemic biomarkers between control and treatment visit. The results are ranked by their p-value, with most significant differences at the top of the list.

Biomarker	Units	control	drug	change (%)
AUC for Cb (0-35min)	mM*sec	45.3 (7.3)	54.0 (12.0)	17.2 (16.0)
AUC for Cb (0-inf)	mM*sec	74.8 (29.0)	110.0 (30.0)	59.5 (42.0)
Body extraction fraction	%	6.81 (2.8)	3.49 (0.67)	-32.1 (29.0)
Cardiac output	L/min	5.07 (1.4)	5.89 (0.8)	29.2 (30.0)
Heart-lung dispersion	%	37.5 (6.2)	46.9 (7.3)	32.0 (31.0)
Heart-lung mean transit time	sec	17.5 (3.2)	17.2 (2.6)	3.48 (21.0)
Organs blood mean transit time	sec	31.1 (8.1)	33.6 (7.2)	15.3 (22.0)
Organs extraction fraction	%	20.4 (3.0)	14.2 (2.6)	-29.9 (7.8)
Organs extravascular mean transit time	min	6.35 (2.3)	7.56 (1.1)	44.6 (50.0)
RE for R1b at 20min	%	17.1 (3.2)	22.7 (5.8)	30.5 (23.0)
RE for Sb at 20min	%	16.2 (6.0)	18.1 (6.6)	40.6 (64.0)

Table 1.5: Mean values along with their 95 percent confidence intervals for all systemic biomarkers at the control and treatment visit. The last column shows the relative change at the treatment visit. The results are ranked by their p-value, with most significant differences at the top of the list.